

ARTIFICIAL INTELLIGENCE GOVERNANCE AND INDIAAI

AI TAXONOMY LADDER → CAPABILITY-TO-DEPLOYMENT LADDER → FOUNDATION-MODEL RISK TREE → INDIAAI INSTITUTION MAP → MISSION PILLAR WHEEL → COMPUTE MEASUREMENT FIREWALL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 AI TAXONOMY LADDER

AI TAXONOMY LADDER ARTIFICIAL INTELLIGENCE BROAD TASK-PERFORMING SYSTEMS MACHINE LEARNING PATTERNS LEARNED FROM DATA DEEP LEARNING MULTILAYER NEURAL NETWORKS GENERATIVE AI

ARTIFICIAL INTELLIGENCE → broad task-performing systems

MACHINE LEARNING → patterns learned from data

DEEP LEARNING → multilayer neural networks

GENERATIVE AI → creates text, image, audio or code

STARTING CONDITION / INPUT

- ARTIFICIAL INTELLIGENCE → broad task-performing systems
- MACHINE LEARNING → patterns learned from data

SCIENTIFIC OR ENGINEERING MECHANISM

- DEEP LEARNING → multilayer neural networks
- GENERATIVE AI → creates text, image, audio or code

OUTPUT / FAILURE BOUNDARY

- RULE → identify method, task and evidence before impact
- ARTIFICIAL INTELLIGENCE → broad task-performing systems

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → identify method, task and evidence before impact.

01

STAGE 01 CAPABILITY-TO-DEPLOYMENT LADDER

CAPABILITY-TO-DEPLOYMENT LADDER RESEARCH CLAIM PROPOSED METHOD BENCHMARK RESULT BOUNDED EVALUATION INTEGRATED DEMONSTRATION REAL WORKFLOW WITH USERS AND MONITORING MEASURED BENEFIT AND HARM AFTER USE

RESEARCH CLAIM → proposed method

BENCHMARK RESULT → bounded evaluation

PROTOTYPE → integrated demonstration

DEPLOYMENT → real workflow with users and monitoring

STARTING CONDITION / INPUT

- RESEARCH CLAIM → proposed method
- BENCHMARK RESULT → bounded evaluation

SCIENTIFIC OR ENGINEERING MECHANISM

- PROTOTYPE → integrated demonstration
- DEPLOYMENT → real workflow with users and monitoring

OUTPUT / FAILURE BOUNDARY

- OUTCOME → measured benefit and harm after use
- RESEARCH CLAIM → proposed method

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME → measured benefit and harm after use.

02

STAGE 02 FOUNDATION-MODEL RISK TREE

FOUNDATION-MODEL RISK TREE PRETRAINED FOUNDATION MODEL BROAD DOWNSTREAM REUSE PROVENANCE CONSTRAINTS INHERITED APPLICATION RISK FLUENT OUTPUT WITHOUT VERIFICATION GROUNDING, EVALUATION AND HUMAN REVIEW

PRETRAINED FOUNDATION MODEL → broad downstream reuse

DATA / LICENCE → provenance constraints

BIAS / SAFETY → inherited application risk

HALLUCINATION → fluent output without verification

CONCEPT / ARCHITECTURE

- PRETRAINED FOUNDATION MODEL → broad downstream reuse
- DATA / LICENCE → provenance constraints

MECHANISM / APPLICATION

- BIAS / SAFETY → inherited application risk
- HALLUCINATION → fluent output without verification

READINESS / STATUS LIMIT

- CONTROL → grounding, evaluation and human review
- PRETRAINED FOUNDATION MODEL → broad downstream reuse

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HALLUCINATION → fluent output without verification; CONTROL → grounding, evaluation and human review.

03

STAGE 03 INDIAAI INSTITUTION MAP

INDIAAI INSTITUTION MAP MINISTRY POLICY AND RULE-MAKING ANCHOR DIGITAL INDIA CORPORATION MISSION IMPLEMENTATION NITI AAYOG EARLIER STRATEGY AND THINK-TANK ROLE SECTOR REGULATOR DEPLOYMENT OBLIGATIONS

MeitY → ministry policy and rule-making anchor

CONCEPT / ARCHITECTURE

- PRETRAINED FOUNDATION MODEL → broad downstream reuse
- DATA / LICENCE → provenance constraints

MECHANISM / APPLICATION

- BIAS / SAFETY → inherited application risk
- HALLUCINATION → fluent output without verification

READINESS / STATUS LIMIT

- CONTROL → grounding, evaluation and human review
- PRETRAINED FOUNDATION MODEL → broad downstream reuse

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HALLUCINATION → fluent output without verification; CONTROL → grounding, evaluation and human review.

03

STAGE 03

INDIAAI INSTITUTION MAP

INDIAAI INSTITUTION MAP

MINISTRY POLICY AND RULE-MAKING ANCHOR

DIGITAL INDIA CORPORATION

MISSION IMPLEMENTATION

NITI AAYOG

EARLIER STRATEGY AND THINK-TANK ROLE

SECTOR REGULATOR

DEPLOYMENT OBLIGATIONS

MeitY → ministry policy and rule-making anchor

INDIAAI / DIGITAL INDIA CORPORATION → mission implementation

NITI AAYOG → earlier strategy and think-tank role

SECTOR REGULATOR / DEPARTMENT → deployment obligations

CONCEPT / ARCHITECTURE

- MeitY → ministry policy and rule-making anchor
- INDIAAI / DIGITAL INDIA CORPORATION → mission implementation

MECHANISM / APPLICATION

- NITI AAYOG → earlier strategy and think-tank role
- SECTOR REGULATOR / DEPARTMENT → deployment obligations

READINESS / STATUS LIMIT

- RULE → strategy, mission, regulator and law stay separate
- MeitY → ministry policy and rule-making anchor

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → strategy, mission, regulator and law stay separate.

04

STAGE 04

MISSION PILLAR WHEEL

MISSION PILLAR WHEEL

COMPUTE + INNOVATION CENTRE + APPLICATIONS

AIKOSH + STARTUP FINANCING + FUTURESILLS

SAFE AND TRUSTED AI

GOVERNANCE TOOLS AND EVALUATION

IMPLEMENTATION DOMAIN

PILLAR NAME IS NOT COMPLETED OUTPUT

COMPUTE + INNOVATION CENTRE + APPLICATIONS

AIKOSH + STARTUP FINANCING + FUTURESILLS

SAFE AND TRUSTED AI → governance tools and evaluation

PILLAR → implementation domain

CONCEPT / ARCHITECTURE

- COMPUTE + INNOVATION CENTRE + APPLICATIONS
- AIKOSH + STARTUP FINANCING + FUTURESILLS

MECHANISM / APPLICATION

- SAFE AND TRUSTED AI → governance tools and evaluation
- PILLAR → implementation domain

READINESS / STATUS LIMIT

- TRAP → pillar name is not completed output
- COMPUTE + INNOVATION CENTRE + APPLICATIONS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → pillar name is not completed output.

05

STAGE 05

COMPUTE MEASUREMENT FIREWALL

COMPUTE MEASUREMENT FIREWALL

GPU OBJECTIVE

PLANNED HARDWARE MEASURE

EMPANELLED CAPACITY

PROCUREMENT AVAILABILITY

COMPUTE UNIT

PROVIDER-DEFINED SERVICE MEASURE

USED CAPACITY

SYSTEM / DEVICE / INSTITUTION

DECISIVE TECHNICAL DISTINCTION

STATUS / UNIT / EVIDENCE LIMIT

GPU OBJECTIVE → planned hardware measure

COMPUTE UNIT → provider-defined service measure

RULE → never convert one measure into another

EMPANELLED CAPACITY → procurement availability

BOOKED / USED CAPACITY → actual access or consumption

GPU OBJECTIVE → planned hardware measure

SYSTEM / DEVICE / INSTITUTION

- GPU OBJECTIVE → planned hardware measure
- EMPANELLED CAPACITY → procurement availability

DECISIVE TECHNICAL DISTINCTION

- COMPUTE UNIT → provider-defined service measure
- BOOKED / USED CAPACITY → actual access or consumption

STATUS / UNIT / EVIDENCE LIMIT

- RULE → never convert one measure into another
- GPU OBJECTIVE → planned hardware measure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → never convert one measure into another.

06

STAGE 06

AIKOSH EVIDENCE GATE

AIKOSH EVIDENCE GATE

DATASET LISTING

LAWFUL ORIGIN AND PERMITTED USE

FITNESS FOR INDIAN CONTEXT

CONTROLLED USE

PLATFORM PRESENCE ALONE PROVES NONE OF THESE

DATASET LISTING → discoverability

PROVENANCE / LICENCE → lawful origin and permitted use

QUALITY / REPRESENTATION → fitness for Indian context

ACCESS / SECURITY → controlled use

OUTCOME → platform presence alone proves none of these

06

SYSTEM / DEVICE / INSTITUTION

- GPU OBJECTIVE → planned hardware measure
- EMPANELLED CAPACITY → procurement availability

DECISIVE TECHNICAL DISTINCTION

- COMPUTE UNIT → provider-defined service measure
- BOOKED / USED CAPACITY → actual access or consumption

STATUS / UNIT / EVIDENCE LIMIT

- RULE → never convert one measure into another
- GPU OBJECTIVE → planned hardware measure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → never convert one measure into another.

06

STAGE 06 AIKOSH EVIDENCE GATE

AIKOSH EVIDENCE GATE

DATASET LISTING

LAWFUL ORIGIN AND PERMITTED USE

FITNESS FOR INDIAN CONTEXT

CONTROLLED USE

PLATFORM PRESENCE ALONE PROVES NONE OF THESE

DATASET LISTING → discoverability

→

PROVENANCE / LICENCE → lawful origin and permitted use

→

QUALITY / REPRESENTATION → fitness for Indian context

→

ACCESS / SECURITY → controlled use

→

OUTCOME → platform presence alone proves none of these

STARTING CONDITION / INPUT

- DATASET LISTING → discoverability
- PROVENANCE / LICENCE → lawful origin and permitted use

SCIENTIFIC OR ENGINEERING MECHANISM

- QUALITY / REPRESENTATION → fitness for Indian context
- ACCESS / SECURITY → controlled use

OUTPUT / FAILURE BOUNDARY

- OUTCOME → platform presence alone proves none of these
- DATASET LISTING → discoverability

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME → platform presence alone proves none of these.

07

STAGE 07 USE-CASE DEPLOYMENT CHAIN

USE-CASE DEPLOYMENT CHAIN

PUBLIC PROBLEM

DEFINE LEGITIMATE OBJECTIVE

DATA + MODEL

BUILD OR PROCURE COMPONENT

INTERFACE + HUMAN WORKFLOW

OPERATIONAL DECISION

MONITORING + APPEAL

PUBLIC PROBLEM → define legitimate objective

→

DATA + MODEL → build or procure component

→

INTERFACE + HUMAN WORKFLOW → operational decision

→

MONITORING + APPEAL → accountability after output

→

RULE → model capability is not service outcome

STARTING CONDITION / INPUT

- PUBLIC PROBLEM → define legitimate objective
- DATA + MODEL → build or procure component

SCIENTIFIC OR ENGINEERING MECHANISM

- INTERFACE + HUMAN WORKFLOW → operational decision
- MONITORING + APPEAL → accountability after output

OUTPUT / FAILURE BOUNDARY

- RULE → model capability is not service outcome
- PUBLIC PROBLEM → define legitimate objective

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → model capability is not service outcome.

08

STAGE 08 RISK LIFECYCLE

RISK LIFECYCLE

PROBLEM SELECTION

EXCLUSION OR MISSION CREEP

BIAS, PRIVACY AND PROVENANCE

HIDDEN FAILURE OR LOCK-IN

AUTOMATION BIAS AND DRIFT

IMPACT ASSESSMENT, LOGS, INCIDENTS AND REVIEW

PROBLEM SELECTION → exclusion or mission creep

→

DATA / TRAINING → bias, privacy and provenance

→

EVALUATION / PROCUREMENT → hidden failure or lock-in

→

DEPLOYMENT / UPDATE → automation bias and drift

→

CONTROL → impact assessment, logs, incidents and review

STARTING CONDITION / INPUT

- PROBLEM SELECTION → exclusion or mission creep
- DATA / TRAINING → bias, privacy and provenance

SCIENTIFIC OR ENGINEERING MECHANISM

- EVALUATION / PROCUREMENT → hidden failure or lock-in
- DEPLOYMENT / UPDATE → automation bias and drift

OUTPUT / FAILURE BOUNDARY

- CONTROL → impact assessment, logs, incidents and review
- PROBLEM SELECTION → exclusion or mission creep

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONTROL → impact assessment, logs, incidents and review.

09

STAGE 09 RISK AND CONTROL MATRIX

RISK AND CONTROL MATRIX

REPRESENTATIVE TESTS AND REMEDY

LAWFUL PURPOSE AND MINIMISATION

ACCESS CONTROL AND RESILIENCE

EVALUATION, EXPLANATION AND ESCALATION

PROVENANCE, LABELLING AND ENFORCEMENT

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
BIAS → representative tests and remedy	SECURITY → access control and resilience	MISINFORMATION → provenance, labelling and enforcement
PRIVACY → lawful purpose and minimisation	SAFETY / OPACITY → evaluation, explanation and escalation	BIAS → representative tests and remedy

SYSTEM / DEVICE / INSTITUTION

- BIAS → representative tests and remedy
- PRIVACY → lawful purpose and minimisation

DECISIVE TECHNICAL DISTINCTION

- SECURITY → access control and resilience
- SAFETY / OPACITY → evaluation, explanation and escalation

STATUS / UNIT / EVIDENCE LIMIT

- MISINFORMATION → provenance, labelling and enforcement
- BIAS → representative tests and remedy

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SAFETY / OPACITY → evaluation, explanation and escalation.

ARTIFICIAL INTELLIGENCE GOVERNANCE AND INDIAAI • TILE 3/4 • same master rows 5005-8239 of 10742 • continues below

09

STAGE 09

RISK AND CONTROL MATRIX

RISK AND CONTROL MATRIX

REPRESENTATIVE TESTS AND REMEDY

LAWFUL PURPOSE AND MINIMISATION

ACCESS CONTROL AND RESILIENCE

EVALUATION, EXPLANATION AND ESCALATION

PROVENANCE, LABELLING AND ENFORCEMENT

SYSTEM / DEVICE / INSTITUTION

BIAS → representative tests and remedy

PRIVACY → lawful purpose and minimisation

DECISIVE TECHNICAL DISTINCTION

SECURITY → access control and resilience

SAFETY / OPACITY → evaluation, explanation and escalation

STATUS / UNIT / EVIDENCE LIMIT

MISINFORMATION → provenance, labelling and enforcement

BIAS → representative tests and remedy

SYSTEM / DEVICE / INSTITUTION

• BIAS → representative tests and remedy

• PRIVACY → lawful purpose and minimisation

DECISIVE TECHNICAL DISTINCTION

• SECURITY → access control and resilience

• SAFETY / OPACITY → evaluation, explanation and escalation

STATUS / UNIT / EVIDENCE LIMIT

• MISINFORMATION → provenance, labelling and enforcement

• BIAS → representative tests and remedy

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SAFETY / OPACITY → evaluation, explanation and escalation.

10

STAGE 10

INSTRUMENT HIERARCHY

INSTRUMENT HIERARCHY

PERSUASIVE COMMUNICATION

CONSULTATION REPORT

NON-BINDING FRAMEWORK

NOTIFIED RULE

BINDING UNDER PARENT ACT

PRIMARY LEGISLATION

SECTORAL LAW

ADVISORY → persuasive communication

CONSULTATION REPORT / GUIDELINE → non-binding framework

NOTIFIED RULE → binding under parent Act

ACT → primary legislation

STARTING CONDITION / INPUT

• ADVISORY → persuasive communication

• CONSULTATION REPORT / GUIDELINE → non-binding framework

SCIENTIFIC OR ENGINEERING MECHANISM

• NOTIFIED RULE → binding under parent Act

• ACT → primary legislation

OUTPUT / FAILURE BOUNDARY

• SECTORAL LAW → context-specific enforceable duty

• ADVISORY → persuasive communication

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ACT → primary legislation; SECTORAL LAW → context-specific enforceable duty.

11

STAGE 11

AI GOVERNANCE ANSWER SPINE

AI GOVERNANCE ANSWER SPINE

METHOD, MODEL, USE CASE AND DEPLOYMENT

MISSION PILLAR, MINISTRY, IMPLEMENTER AND REGULATOR

DATA, MODEL, PROCUREMENT, DECISION AND MONITORING

HUMAN REASONS, CONTESTABILITY AND REMEDY

OUTLAY, COMPUTE, MODEL AND SUMMIT STATUS OFFICIALLY

DEFINE → method, model, use case and deployment

MAP → mission pillar, ministry, implementer and regulator

TRACE → data, model, procurement, decision and monitoring

RETAIN → human reasons, contestability and remedy

VERIFY → outlay, compute, model and summit status officially

CONCEPT / ARCHITECTURE

• DEFINE → method, model, use case and deployment

• MAP → mission pillar, ministry, implementer and regulator

MECHANISM / APPLICATION

• TRACE → data, model, procurement, decision and monitoring

• RETAIN → human reasons, contestability and remedy

READINESS / STATUS LIMIT

• VERIFY → outlay, compute, model and summit status officially

• DEFINE → method, model, use case and deployment

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERIFY → outlay, compute, model and summit status officially.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

INDIA'S AI BOTTLENECK IS NOT ONLY RESEARCH QUALITY

IT IS ALSO COMPUTE ACCESS, MULTILINGUAL DATA, EVALUATION FRAMEWORKS

A PRO-INNOVATION STANCE DOES NOT MEAN REGULATORY VACUUM

IT USUALLY MEANS GUIDELINES, STANDARDS, INSTITUTIONAL OVERSIGHT AND SECTOR-SENSITIVE

INDIA'S GOVERNANCE CHALLENGE IS SHAPED BY SCALE, MULTILINGUALITY, WELFARE-STATE

CENTRAL MINISTRY ANCHOR.

INDIAAI MISSION

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

• CAUTION: India's AI bottleneck is not only research quality

• it is also compute access, multilingual data, evaluation frameworks and diffusion into public-interest use cases.

• CAUTION: A pro-innovation stance does not mean regulatory vacuum

• it usually means guidelines, standards, institutional oversight and sector-sensitive calibration.

READINESS, UNIT OR STATUS LIMIT

• CAUTION: India's governance challenge is shaped by scale, multilinguality, welfare-state interaction and uneven digital capacity.

• MeitY: central ministry anchor.

• IndiaAI Mission: Cabinet-approved mission structure.

• Digital India AI initiative: broad public-facing programme explanation.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

• India AI Governance Guidelines page: official mission-linked governance framing.

• Safe & Trusted AI pillar: official IndiaAI governance/safety pillar.

• AI can strengthen public services, agriculture, health, education, citizen interfaces and productivity tools.

• Indigenous compute and model-development capacity matter for strategic autonomy in language technologies and sensitive sectors.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.